Generalised Dihedral Angle

I. Rouse

September 30, 2026

1 First section

Given three successive bonds between four atoms in a chain, the dihedral angle is defined as the angle between the plane formed by atoms $1, 2, 3$ and $2, 3, 4$, or equivalently the planes defined by the bond vectors $u = 1, 2; v = 2, 3; w = 3, 4$, i.e. the plane uv and vw. In $\mathcal{R}^3$, a relatively simple formula for this angle is given by,

$$\cos\phi = \frac{(u \times v) \cdot (v \times w)}{|u \times v||v \times w|} \tag{1}$$

$$\sin\phi = \frac{v \cdot ((u \times v) \times (v \times w))}{|v||u \times v||v \times w|} \tag{2}$$

where $\times$ is the vector cross product. In practice, the atan2 function is typically used to evaluate the angle by combining the above two expressions to ensure the correct sign is returned.

In the above, the use of the vector cross product is used to produce a vector normal to each plane defined by pairs of bonds, since the angle between these vectors is equal to the angle between the planes. In dimensions other than 3, the vector cross product is not uniquely defined and so this definition cannot be directly extended to higher dimensional spaces. We thus must turn to the mathematical formalism of bivectors. Given two vectors u, v we may consider the geometric product uv as the non-commutative product of their expansions in terms of components u_i and basis vectors e_i:

$$uv = (u_1e_1 + u_2e_2 + ...)(v_1e_1 + v_2e_2....) \tag{3}$$

$$uv = u_1e_1v_1e_1 + u_1e_1v_2e_2 + u_2e_2v_1e_1 + u_2e_2v_2e_2 + ... \tag{4}$$

The products of basis vectors e_ie_j have the property that $e_i^2 = 1$ and otherwise $e_ie_j = -e_je_i$, thus:

$$uv = u_1v_1 + u_1v_2e_1e_2 - u_2v_1e_1e_2 + u_2v_2 + ... \tag{5}$$

and likewise

$$vu = u_1v_1 - u_1v_2e_1e_2 + u_2v_1e_1e_2 + u_2v_2 + ... \tag{6}$$

The geometric product is associative, distributitive, and further has the property that $uu = u^2 = |u^2|$ is scalar valued for any u, as can be seen from the above.

Let us first consider the symmetric combination, $uv + vu$:

$$uv+vu = u_1v_1+u_1v_2e_1e_2-u_2v_1e_1e_2+u_2v_2+.u_1v_1-u_1v_2e_1e_2+u_2v_1e_1e_2+u_2v_2+... \quad (7)$$

$$uv + vu = 2u_1v_1 + 2u_2v_2 + ... \quad (8)$$

which we may immediately see gives us the vector dot product,

$$u \cdot v = \frac{1}{2}(uv + vu) \quad (9)$$

We may instead consider the antisymmetric product, which we denote $u \wedge v$:

$$u \wedge v = \frac{1}{2}(uv - vu) \quad (10)$$

where for example

$$uv-vu = u_1v_1+u_1v_2e_1e_2-u_2v_1e_1e_2+u_2v_2+...-(u_1v_1-u_1v_2e_1e_2+u_2v_1e_1e_2+u_2v_2+...) \quad (11)$$

such that:

$$uv - vu = 2u_1v_2e_1e_2 - 2u_2v_1e_1e_2 + ... \quad (12)$$

which is clearly not a regular vector or scalar, but instead represented in terms of products of vector components, and is a first example of a bivector. A simple bivector $\mathbf{A}$ is defined as the exterior product between two standard vectors, $\mathbf{A} = u \wedge v = -v \wedge u$, and represents the directed area of the plane defined by u, v, with magnitude $|\mathbf{A}| = |u||v| \sin\theta_{uv}$, where θ_{uv} is the angle between the two vectors.

The geometric product of two bivectors can be expressed in the general form,

$$\mathbf{AB} = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \times \mathbf{B} + \mathbf{A} \wedge \mathbf{B} \quad (13)$$

where $\mathbf{A} \cdot \mathbf{B} = \frac{1}{2}(AB + BA)$ is the scalar-valued dot product, $\mathbf{A} \times \mathbf{B} = \frac{1}{2}(AB - BA)$ is a bivector-valued commutator product, and $\mathbf{A} \wedge \mathbf{B}$ is a 4-vector valued exterior product. In three dimensions, the relations $|\mathbf{A} \cdot \mathbf{B}| = |A||B| \cos\theta$ and $\mathbf{A} \times \mathbf{B} = |A||B| \sin\theta$ provide a means to extract the angle between the planes defined by $\mathbf{A}, \mathbf{B}$, analogously to the use of the vector cross product on u, v, w, but with the advantage that these directly extend to higher dimensional spaces.

In the particular case here, we define the bivectors $\mathbf{A} = u \wedge v, \mathbf{B} = v \wedge w, \mathbf{C} = u \wedge w$, and we are interested in the angle between bivectors $\mathbf{A}, \mathbf{B}$. We begin by writing down the geometric products for this pair,

$$\mathbf{AB} = (u \wedge v)(v \wedge w) \quad (14)$$

Substituting in the definitions for the wedge product, we have:

$$\mathbf{AB} = \frac{1}{4}(uv - vu)(vw - wv) \quad (15)$$

and expansion gives,

$$\mathbf{AB} = \frac{1}{4}(uvvw - vuvw - uvwv + vuwv) \tag{16}$$

To simplify this expression, we make use of the associative property of geometric products, $vuvw = (vu)(vw) = v(uv)w$, the property $vv = |v^2|$, and the relation $uv + vu = 2u \cdot v \rightarrow uv = 2u \cdot v - vu$ to convert the above four four-vector combinations into forms in which both v components are neighbouring to allow these to be reduced to a scalar:

$$uvvw = |v^2|uw \tag{17}$$

$$vuvw = (2u \cdot v - uv)vw = 2(u \cdot v)vw - |v^2|uw \tag{18}$$

$$uvwv = uv(2v \cdot w - vw) = 2(v \cdot w)uv - |v^2|uw \tag{19}$$

$$vuwv = (2u \cdot v - uv)(2v \cdot w - vw) = 4(u \cdot v)(v \cdot w) - 2(v \cdot w)uv - 2(u \cdot v)vw + |v^2|uw \tag{20}$$

Such that finally:

$$\mathbf{AB} = |v^2|uw + (u \cdot v)(v \cdot w) - (v \cdot w)uv - (u \cdot v)vw \tag{21}$$

in which each component is now either a scalar or the geometric product of two vectors. Repeating the same procedure with the product $\mathbf{BA}$:

$$\mathbf{BA} = \frac{1}{4}(vwuv - wvuv - vwvu + wvvu) \tag{22}$$

$$vwuv = (2v \cdot w - wv)(2u \cdot v - vu) = 4(v \cdot w)(u \cdot v) - 2(v \cdot w)vu - 2(u \cdot v)wv + |v^2|wu \tag{23}$$

$$wvuv = wv(2u \cdot v - vu) = 2(u \cdot v)wv - |v^2|wu \tag{24}$$

$$vwvu = (2v \cdot w - wv)vu = 2(v \cdot w)vu - |v^2|wu \tag{25}$$

$$wvvu = |v^2|wu \tag{26}$$

$$\mathbf{BA} = |v^2|wu + (u \cdot v)(v \cdot w) - (v \cdot w)vu - (u \cdot v)wv \tag{27}$$

Unsurprisingly, this has the same general form as $\mathbf{AB}$ but the opposite order for the remaining vector products. In both cases, there is no 4-vector term as a result of the shared vector between $\mathbf{A}$ and $\mathbf{B}$. The scalar term is given by,

$$\mathbf{A} \cdot \mathbf{B} = 1/2[\mathbf{AB} + \mathbf{BA}] \tag{28}$$

$$\mathbf{A} \cdot \mathbf{B} = 1/2(|v^2|(uw + wu) + 2(u \cdot v)(v \cdot w) - (v \cdot w)[uv + vu] - (u \cdot v)[vw + wv]) \tag{29}$$

and using the definition for the vector dot products we have,

$$\mathbf{A} \cdot \mathbf{B} = |v^2|(u \cdot w) - (u \cdot v)(v \cdot w) \tag{30}$$

For consistency we require that $\mathbf{A}\cdot\mathbf{B} = |A||B|\cos\phi$, where ϕ is the required dihedral angle. Thus:

$$\cos\phi = \frac{|v^2|(u\cdot w) - (u\cdot v)(v\cdot w)}{|u||v|\sin\theta_{uv}|v||w|\sin\theta_{vw}} \tag{31}$$

As a reminder, the 3D vector cross product between u and v has the same magnitude $|u||v|\sin\theta_{uv}$ as the bivector generated by $u\wedge v$ and so this is equivalent to:

$$\cos\phi = \frac{|v^2|(u\cdot w) - (u\cdot v)(v\cdot w)}{|u\times v||v\times w|} \tag{32}$$

To confirm that the numerator is consistent with the standard form, we may use the Binet-Cauchy identity,

$$(a\times b)\cdot(c\times d) = (a\cdot c)(b\cdot d) - (a\cdot d)(b\cdot c) \tag{33}$$

which in this case gives

$$(u\times v)\cdot(v\times w) = (u\cdot v)(v\cdot w) - (u\cdot w)|v^2| \tag{34}$$

$$\mathbf{A}\times\mathbf{B} = 1/2[\mathbf{AB} - \mathbf{BA}] \tag{35}$$

The scalar term can be straighforwardly evaluated using efficient computational routines, since it involves only scalar products between standard vectors.